Milestone 3 - UI Document

Team: Team 01 - SmashGo

Topic: D2 - Sports Platform

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Repository: https://github.com/SE-FDA-NEU-AI66B/software-group1-topicd2

Project board: https://github.com/orgs/SE-FDA-NEU-AI66B/projects/24

Setup guide: https://github.com/SE-FDA-NEU-AI66B/software-group1-topicd2/blob/main/docs/SETUP.md

Submitted by: Chử Vũ Thảo Hiền

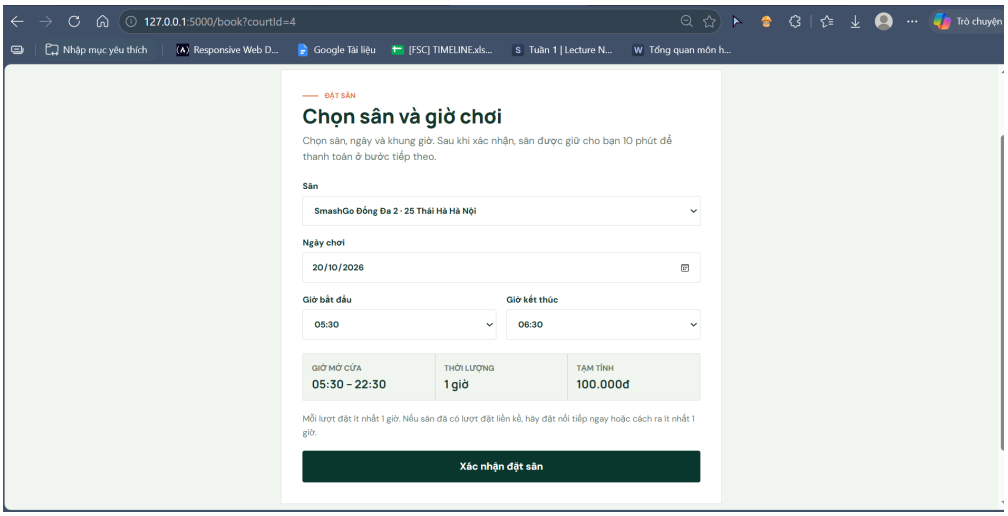
Project board right after Sprint 2 Planning:



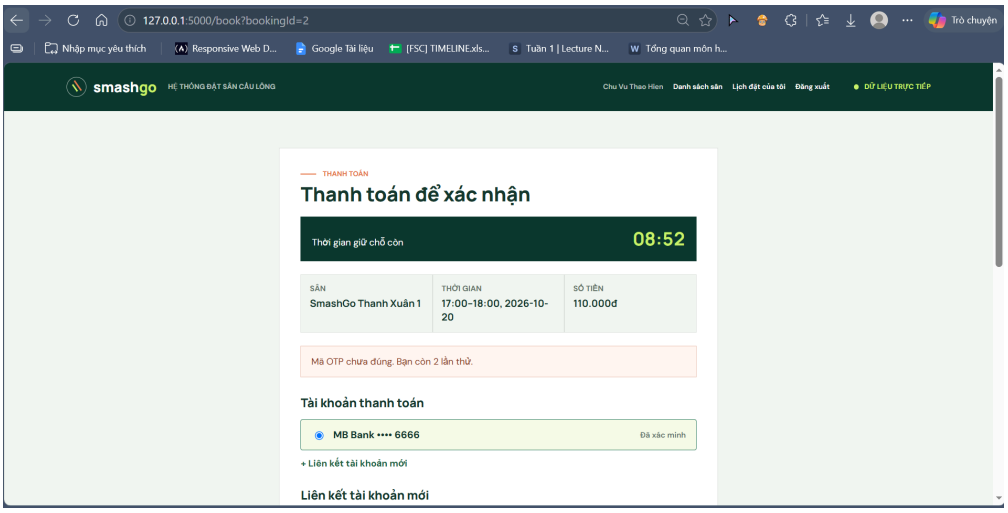
Project board on the submission day:



P0 screen with real data



Error message shown to the user



UI Screens and Flows

This document maps user stories to the screens and routes currently implemented in the application. P0 screens have a wireframe in `docs/images/ui/`.

1. P0 stories and status

The M1 backlog marks US-01 to US-05 and US-11 to US-12 as P0. Status values use the required `works`, `partly works`, or `not started` labels.

Story	Screen(s)	Route	Status	Issue
US-01 — View available courts	Court list and filters	`GET /courts` (`/` redirects here)	works	#95 — court discovery and availability
US-02 — View court details	Court details	`GET /courts/{courtId}`	works	#95 — court discovery and availability

Story	Screen(s)	Route	Status	Issue
US-03 — View available time slots	Court details and availability	`GET /courts/{courtId}`	works	#95 — court discovery and availability
US-04 — Book one available time slot	Booking; login or registration; payment; my bookings	`GET /book`; `GET /login`; `GET /register`; `GET /book?bookingId={bookingId}`; `GET /my-bookings`	works	#91 — booking hold; #92 — payment; #96 — authentication
US-05 — Prevent double booking	Court availability and booking	`GET /courts/{courtId}`; `GET /book`	works	#91 — booking hold
US-11 — Activate or deactivate courts	Admin court management	`GET /admin/courts`	works	#94 — Admin court management
US-12 — Approve or reject court request	Admin court management	`GET /admin/courts`	works	#94 — Admin court management

Updated screen flow from M1

Guest and User booking flow

```
GET / or /courts
  -> filter courts by location/date/time
  -> GET /courts/{courtId}
       -> GET /api/courts/{courtId}
       -> GET /api/courts/{courtId}/availability?date=...
  -> choose an available slot
  -> GET /book?courtId=...&date=...&startTime=...&endTime=...
       -> if Guest: GET/POST /login or GET/POST /register, then return to /book
  -> POST /api/bookings
       -> create a 10-minute PENDING hold
  -> GET /book?bookingId={bookingId}
       -> link bank account if needed, then verify the account with OTP
       -> request a payment OTP for the selected verified account
       -> enter the payment OTP
       -> POST /api/bookings/{bookingId}/payment
  -> GET /my-bookings
```

An authenticated User skips sign-in and registration. `/` is a redirect to `/courts`, not a separate screen. The M1 `/availability` screen is replaced by the filters on `/courts` and date-specific availability on

`/courts/{courtId}`; `/availability` is not a current application route.

Admin flow

```
GET/POST /login
  -> role = ADMIN
  -> GET /admin/courts
       -> GET /api/admin/court-requests
       -> POST /api/admin/court-requests/{requestId}/review
       -> GET /api/admin/courts
       -> PATCH /api/admin/courts/{courtId}/status
```

Manager flow

```
GET/POST /login
  -> role = MANAGER
  -> GET /manager/courts
       -> P1 route; the page is currently an unsupported-feature placeholder
```

2. Wireframes

Screen(s)	"The user comes here to ..."	API endpoints	Wireframe
Court list, details, and availability	find courts, review court details, and choose an available time slot.	`GET /api/courts`; `GET /api/courts/{courtId}`; `GET /api/courts/{courtId}/availability?date=...`	images/ui/courts-availability.svg
Booking and hold creation	choose a court, date, and time, then create a temporary booking hold.	`POST /api/bookings`	images/ui/book.svg
Login and registration	sign in or create a User	No JSON API; form submissions: `POST /login`, `POST /register`	images/ui/auth.svg

Screen(s)	"The user comes here to ..."	API endpoints	Wireframe
	account and return to the requested booking screen.		
Payment, bank account, and OTP	link and verify a bank account, request a payment OTP, and pay for a held booking.	`GET /api/bank-accounts`; `POST /api/bank-accounts`; `POST /api/bank-accounts/{bankAccountId}/verify`; `POST /api/bookings/{bookingId}/payment/otp`; `POST /api/bookings/{bookingId}/payment`	images/ui/payment.svg
My bookings	view booking status and details, and manage eligible bookings.	`GET /api/bookings/mine`; `DELETE /api/bookings/{bookingId}` (P1 cancellation)	images/ui/my-bookings.svg
Admin court review and status management	review new court requests and activate or deactivate courts.	`GET /api/admin/court-requests`; `POST /api/admin/court-requests/{requestId}/review`; `GET /api/admin/courts`; `PATCH /api/admin/courts/{courtId}/status`	images/ui/admin-courts.svg

3. Error messages

API errors use the response shape `{ "code": "...", "message": "..." }`. The screen presenting an actionable error should preserve the form values and show the message as an alert. These M2 messages are the required user-facing copy:

HTTP status	Code	Rule	Message
409	`SLOT_UNAVAILABLE`	BR1	"This court was just booked for that time. Please choose another time or court."
409	`ACTIVE_BOOKING_LIMIT`	BR6	"You already have 2 active bookings. Complete or cancel one before booking another."
422	`BOOKING_RULE_VIOLATION`	BR2	"A booking must last at least 1 hour; bookings on the same court must be back-to-back or at least 1 hour apart."
422	`BANK_ACCOUNT_NOT_VERIFIED`	BR7	"Your bank account is not verified yet. Enter the OTP to verify it before paying."
409	`HOLD_EXPIRED_OR_ALREADY_PAID`	BR3	"The 10-minute hold has ended and the slot was released. Please book again."

`SLOT_UNAVAILABLE`, `ACTIVE_BOOKING_LIMIT`, and `BOOKING_RULE_VIOLATION` are returned by `POST /api/bookings`. `BANK_ACCOUNT_NOT_VERIFIED` and `HOLD_EXPIRED_OR_ALREADY_PAID` are returned by `POST /api/bookings/{bookingId}/payment`. When the 10-minute payment hold expires, the booking status is `EXPIRED` and the slot is released.

4. What changed since M2

M2 documented the architecture and API contracts. Since then, the repository has added the corresponding user-facing routes, screens, API handlers, and automated tests:

Change 1: Designed court discovery and booking flows are implemented

- **What changed:** Court search, court details and availability now lead into sign-in or registration when needed, a 10-minute booking hold, and payment. Booking creation checks overlapping bookings within a serialized `BEGIN IMMEDIATE` transaction.
- **Evidence:** `routes/courts.py`, `routes/auth.py`, `routes/bookings.py`, `booking_service.py`, and `tests/test_courts.py` / `tests/test_booking.py`.
- **Effect on this document:** The current routes and their implemented status are listed above. `/availability` is not a standalone route; filters are on `/courts` and availability is shown on `/courts/{courtId}`.

Change 2: Payment now requests a separate OTP for each payment

- **What changed:** In addition to verifying a linked bank account, the payment flow sends a payment OTP to the selected verified account before charging it. The first linked account is marked primary and selected by default. The payment OTP expires after five minutes and is limited to three attempts.
- **Evidence:** `POST /api/bookings/{bookingId}/payment/otp` in `routes/bookings.py`; OTP and primary-account handling in `payment_service.py` and `static/payment.js`; coverage in `tests/test_payment.py`.
- **Effect on this document:** The payment flow includes requesting and entering a payment OTP before `POST /api/bookings/{bookingId}/payment`. Only the account's last four digits are displayed.

Change 3: Booking management and Admin screens are implemented

- **What changed:** Users can view their bookings and cancel eligible bookings; Admins can review court requests and activate or deactivate courts. The Manager court workspace is still an unsupported-feature placeholder and is not implemented.
- **Evidence:** `routes/bookings.py` and `booking_service.py`; `routes/admin.py` and `admin_service.py`; corresponding tests in `tests/test_my_bookings.py` and `tests/test_admin_courts.py`.
- **Effect on this document:** `/my-bookings` and `/admin/courts` are marked Implemented in the table above; `/manager/courts` remains Placeholder.